future work to explore new intrinsic/upstream surrogate measures of performance that can be shown to reliably predict downstream results (including for desiderata beyond accuracy) as discussed in Bommasani et al. (2021, §4.4.2)

25. **Trends for model scale.** We find that model scale, within a model family, reliably predicts model accuracy, but for no scenario is a good predictor of downstream accuracy across all models (Figure 29). However, we see a very clear thresholding effect: all models that win head-to-head model comparisons for accuracy at a rate well above chance (i.e. $> 55\%$) are at least 50B parameters (Figure 26). Of these models, which are the 10 most accurate models, some of the most accurate (i.e. in the top 5) are the smallest (Anthropic-LM v4-s3 (52B), Cohere xlarge v20220609 (52.4B)). Overall, scale seems to be a key determinant of accuracy, and scaling within a model family reliable improves accuracy, but it might be inefficient compared to other means (e.g. training with human feedback; compare TNLG v2 (530B) and Anthropic-LM v4-s3 (52B)).

1.3 Contributions

To summarize, our contributions are:

1. **Taxonomy.** We taxonomize the vast design space of language model evaluation into scenarios and metrics. By stating this taxonomy, we can select systematically from this space, which makes explicit both *our priorities* in benchmark design and *the limitations* in the benchmark at present (see §10: MISSING).

2. **Broad coverage.** Given our taxonomy, we select and implement 16 core scenarios, for which we comprehensively measure 7 metrics (accuracy, calibration, robustness, fairness, bias, toxicity, efficiency). We also include 7 targeted evaluations of skills and risks (e.g. knowledge, reasoning, disinformation, copyright), introducing 21 new scenarios that have not been previously used in mainstream language model evaluation.

3. **Evaluation of existing models.** We evaluate 30 language models under the standardized conditions of our benchmark, ensuring models can now be directly compared across many scenarios and metrics. These models vary in terms of their public accessibility: 10 are open, 17 are limited-access, and 3 are closed.

4. **Empirical findings.** Our extensive evaluation yields a host of findings (§8: EXPERIMENTS), which in some cases reinforce findings in the literature and in others produce new knowledge about today's language models. These results offer guidance for future language model development and ample opportunities for further analysis.

5. **Interactive results and codebase.** We provide a public website with all results, underlying model predictions and adaptation details, along an extensible codebase to support the community in taking HELM further.[28]

Acknowledging the prior work this effort builds on. To build our holistic evaluation of language models, we directly build on top of many prior works. While we advocate for evaluating language models in their totality, i.e. centralizing many disparate evaluations, we want to be explicit that the underlying works across the AI community **should be recognized and cited**, as HELM would not exist in its current form without them. In particular, if the results of HELM are used by future work or new models are evaluated on HELM, they should cite the works that created the many datasets/evaluations that constitute HELM.[29] For this reason, we provide the BibTeX entries for all of these works in the codebase[30] and explicitly acknowledge the associated work for every evaluation on the website.[31]

[28] `https://crfm.stanford.edu/helm/v0.1.0`

[29] We take direct inspiration from and follow the precedent set by GLUE and SuperGLUE (Wang et al., 2019a;b); see `https://super.gluebenchmark.com/faq`.

[30] `https://github.com/stanford-crfm/helm`

[31] `https://crfm.stanford.edu/helm/v0.1.0`

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